Fixed-Base Digital Arithmetic via Orbital Angular Momentum Spatial Shifting

Introduction to Optical Digital Computing Architecture

The historical trajectory of computing architectures has been predominantly defined by the evolution of electronic semiconductor devices, governed for decades by the empirical predictions of Moore's Law and Dennard scaling. However, as transistor gate lengths approach fundamental thermodynamic and quantum mechanical limits—most notably governed by Landauer's principle regarding the minimum energy required to erase a bit of information—the constraints of the von Neumann bottleneck have become critically pronounced. The energy dissipation and inherent latency associated with shuttling electrons between physically separated memory and processing units necessitate the exploration of alternative, non-electronic computational substrates. Optical computing has emerged as a premier candidate to transcend these limitations, leveraging the ultra-high bandwidth, extreme parallelism, inherent multiplexing capabilities, and near-zero energy consumption of passive photon propagation[1].

While analog optical computing has seen significant advances in specialized tasks such as Fourier transforms, spatial filtering, and matrix-vector multiplications, the realization of general-purpose optical computers remains strictly contingent upon the development of robust, scalable all-optical digital logic and arithmetic architectures. The primary challenge lies in executing precise mathematical operations without relying on power-hungry optical-to-electrical-to-optical (O-E-O) conversions or highly sensitive optical non-linearities, which require immense power densities to trigger[1].

Recent advancements in nanophotonics and structured light have identified the orbital angular momentum (OAM) of light as a pivotal degree of freedom for encoding multidimensional information. Unlike intrinsic spin angular momentum (SAM), which is physically constrained to two orthogonal states corresponding to left- and right-circular polarization, OAM possesses a theoretically infinite state space[2]. By structuring the phase front of a laser beam into a helical shape, photons can be imbued with a quantized topological charge. Concurrently, the advent of Optical Diffractive Neural Networks (ODNNs) has provided a highly flexible framework for manipulating complex optical fields. Composed of cascaded layers of passive phase and amplitude modulators, ODNNs can be trained *in silico* using deep learning algorithms and deployed as physical optical hardware to perform specific tasks at the speed of light[4].

A groundbreaking synthesis of these domains is the utilization of ODNNs to perform fixed-base digital arithmetic through the spatial shifting of OAM modes. Rather than relying on continuous phase alterations to directly scale the topological charge of a single beam, this paradigm exploits the spatial positioning of independent OAM beams to represent the magnitude of a digit in a positional numeral system. This maps arithmetic multiplication and division directly onto the physical left and right spatial shifting of these optical modes across a transverse

diffractive plane[4]. The subsequent analysis provides an exhaustive examination of this architecture, detailing the theoretical quantum physics of OAM, the diffraction mechanics of ODNNs, the mathematical mapping of fixed-base arithmetic onto spatial coordinates, alternative geometric coordinate transformations, and the operational robustness of these systems under physical and atmospheric perturbations.

Theoretical Foundations of Angular Momentum in Optical Systems

Spin and Orbital Angular Momentum Kinematics

At the fundamental level, atoms and photons engage in complex interactions governed by stringent conservation laws involving energy, linear momentum, and total angular momentum. Particles possess two forms of angular momentum: intrinsic spin angular momentum and extrinsic orbital angular momentum related to spatial distribution[2]. The mathematical formulation of angular momentum relies heavily on the theory of rotation matrices and group generators. Just as linear momentum acts as the generator for spatial translations, angular momentum operators serve as the generators of rotations in a given space[7].

The commutation relations among the components of the orbital angular momentum vector $\mathbf{L} = (L_x, L_y, L_z)$ form the bedrock of this framework. Operating in a quantum mechanical or structured wave context, the differential operators representing these generators in spherical coordinates are mathematically defined to exhibit specific non-commutative algebra:

$$[L_i, L_j] = i\epsilon_{ijk} L_k$$

where ϵ_{ijk} is the Levi-Civita symbol. For spatial wavefronts, such as the cross-section of a structured light beam, the azimuthal derivative dictates the angular momentum along the propagation axis[6]. The operator L_z is formulated as $L_z = -i\frac{\partial}{\partial \phi}$. The eigenstates of this operator, corresponding to the azimuthal phase dependence, are characterized by eigenvalues m (or topological charge l in optics). Consequently, the angular behavior of a particle or a photon in a given state ψ is intrinsically tied to these spatial wave functions[7].

Spin-Orbit Interaction and Spatial Deformations

Understanding the coupling between spin and orbital angular momentum is critical for maintaining the integrity of computational optical beams. The total angular momentum $\mathbf{J}$ is the vector sum of orbital ($\mathbf{L}$) and spin ($\mathbf{S}$) angular momenta. In isotropic free space, measuring one does not disturb the other, a mathematical fact embodied by the vanishing

commutator $[L_z, S_x] = 0$, indicating that orbital and spin degrees of freedom are independent and non-interfering[6].
However, in highly focused beams, sub-wavelength optical elements, or inhomogeneous media, spin-orbit coupling manifests. The interaction energy is proportional to the dot product $\mathbf{L} \cdot \mathbf{S}$, which implies that $\mathbf{L}$ and $\mathbf{S}$ are no longer perfectly independent[6]. The squaring of the total angular momentum operator yields $\mathbf{J}^2 = \mathbf{L}^2 + \mathbf{S}^2 + 2\mathbf{L} \cdot \mathbf{S}$. This coupling exerts a minute torque on the orbit, leading to the physical spatial shifting of spin-polarized light beams[8]. For precision optical arithmetic systems, these inherent spin-orbit shifts must be anticipated, as unintended spatial deviations can map to erroneous digital bit shifts, corrupting the computational output.

Wave Mechanics of Laguerre-Gaussian Beams

The practical generation of OAM relies on solutions to the paraxial Helmholtz equation. The Laguerre-Gaussian (LG) mode is the most frequently utilized spatial profile. The complex amplitude of an LG beam contains the azimuthal phase term $\exp(il\phi)$, where the topological charge l defines the number of intertwined helices in the phase front[3].
The amplitude distribution of an LG beam at its beam waist ($z = 0$) in cylindrical coordinates (r, ϕ, z) is characterized rigorously by:

$$u_{p,l}(r, \phi) = \sqrt{\frac{2p!}{\pi(p + |l|)!}} \frac{1}{w_0} \left(\frac{r\sqrt{2}}{w_0} \right)^{|l|} L_p^{|l|} \left(\frac{2r^2}{w_0^2} \right) \exp\left(-\frac{r^2}{w_0^2} \right) \exp(il\phi)$$

where w_0 is the beam waist radius, $L_p^{|l|}$ represents the generalized Laguerre polynomial, and p is the radial index defining the number of concentric intensity rings. Because the phase is singular at $r = 0$, destructive interference creates a central optical vortex, appearing as a dark hollow core[3]. The orthogonality of these modes over the azimuthal integral allows an infinite set of OAM states to function as a high-dimensional spatial basis for information encoding[3].

Optical Diffractive Neural Networks as Computational Substrates

Diffraction Mechanics and the Rayleigh-Sommerfeld Formulation

Optical diffractive neural networks represent a paradigm shift from traditional artificial neural networks (ANNs). Instead of relying on electronic multiply-accumulate (MAC) units, an ODNN implements matrix multiplications and non-linear feature extractions physically as continuous light waves propagate through space and modulate across diffractive optical elements (DOEs)[4]. An ODNN architecture is composed of a series of spatially separated layers, each comprising thousands of sub-wavelength or micrometer-scale discrete pixels that independently modulate the amplitude and/or phase of the incident electromagnetic field[4]. The forward propagation of light between consecutive layers in an ODNN is modeled mathematically by the Rayleigh-Sommerfeld diffraction integral. If a complex field $U_i(x_i, y_i)$ is transmitted through the i-th diffractive layer, the incident field $U_{i+1}(x_{i+1}, y_{i+1})$ at the subsequent layer located at an axial distance z is computed using the spatial impulse response (or free-space point spread function):

$$h(x, y, z) = \frac{1}{j\lambda} \frac{z}{r^2} \exp(jkr)$$

where $r = \sqrt{x^2 + y^2 + z^2}$, λ is the operational wavelength, and $k = 2\pi/\lambda$ is the wave number. The field arriving at layer $i + 1$ is the two-dimensional convolution of the modulated field at layer i with this impulse response:

$$U_{i+1}(x_{i+1}, y_{i+1}) = \iint [U_i(x_i, y_i) \cdot t_i(x_i, y_i)] \, h(x$$

Here, $t_i(x_i, y_i) = A_i(x_i, y_i) \exp[j\phi_i(x_i, y_i)]$ dictates the complex transmission function of the i-th layer. To maximize optical throughput, energy efficiency, and operational simplicity, phase-only ODNNs are typically preferred. In these architectures, the amplitude transmission coefficient is uniformly set to $A_i = 1$, and the network is optimized exclusively by modulating the phase shifts $\phi_i \in [0, 2\pi)$ across the spatial grid[3].

In Silico Training and Phase Quantization

The optimization of the ODNN phase matrices relies on simulating the physical diffraction process *in silico* and utilizing standard deep learning backpropagation algorithms. The training

loop consists of information forward propagation, loss calculation against an ideal target output, error backpropagation through the diffractive layers, and subsequent weight (phase) updates[5]. Algorithms such as Adaptive Moment Estimation (Adam) are employed to minimize loss functions (e.g., Mean Squared Error or categorical cross-entropy) within computational environments equipped with high-performance graphics processing units[5].

Upon the convergence of the loss function, the theoretical continuous phase profiles must be discretized for physical fabrication or dynamic display on spatial light modulators (SLMs). Phase quantization introduces an element of physical error into the ideal mathematical model. Analyses confirm that while binary phase modulation introduces significant modal degradation, phase quantization at or above 4-bit levels (16 discrete phase steps) provides sufficient degrees of freedom to accurately execute OAM spatial shifting operations without catastrophic fidelity loss[4]. This relatively low-bit requirement is highly advantageous for physical implementations.

Energy Efficiency and Computational Complexity (FLOPs)

A primary motivation for replacing electronic arithmetic units with optical diffractive networks is the theoretical limit of computational energy efficiency. To properly quantify this, the total number of floating-point operations (FLOPs) required to simulate the ODNN electronically must be compared against the passive optical execution.

For electronic phase shift neural networks, the total FLOPs for the convolution layers depend heavily on the summation of multiplication and addition operations scaled by the input image height (H), width (W), and kernel dimensions. In stark contrast, the physical ODNN executes the equivalent of a two-dimensional fast Fourier transform (2D FFT) instantaneously during propagation[9]. The mathematical complexity of calculating the magnitude, computing the phase angle, and normalizing the complex operations for a 2D FFT scales as $\mathcal{O}(N \log N)$ where $N = H \times W$ [9]. In the optical domain, the "FLOPs" executed by the diffractive mixing of wavefronts require zero active energy consumption; the only energy expended is the initial laser generation and the static power maintaining the state of the SLMs or the structural integrity of the metasurface. This physically parallel and selective structure-preserving implementation dwarfs electronic back-ends in energy efficiency, adaptability, and scalability[4].

Mathematical Mapping of Fixed-Base Arithmetic to Spatial Shifting

Positional Numeral Systems and Base Multiplication

The core innovation of utilizing ODNNs for digital arithmetic lies in equating the fundamental mathematical operations of integer multiplication and division with the physical spatial shifting of discrete data points across a transverse plane[4]. To construct this framework, numerical data must be represented in a fixed-base positional numeral system.

In a binary numeral system (base $N = 2$), any decimal integer X_D can be uniquely

represented by an ordered sequence of bits $b_i \in \{0, 1\}$, where each discrete position corresponds to a specific exponential weight. The weights increase sequentially from right to left as powers of 2. The decimal equivalent is defined by the polynomial:

$$X_D = b_{n-1}2^{n-1} + b_{n-2}2^{n-2} + \cdots + b_1 2^1 + b_0 2^0$$

In this standard notation, b_0 represents the least significant bit (LSB) and b_{n-1} represents the most significant bit (MSB). By shifting the entire binary sequence uniformly to the left by k bits—and appending zeros to the newly vacated lower-order positions—a new binary sequence is generated. This new sequence corresponds to a modified decimal value X'_D:

$$X'_D = b_{n-1}2^{n+k-1} + \cdots + b_0 2^k$$

By factoring out the term 2^k, the mathematical relationship simplifies to $X'_D = X_D \times 2^k$ [4]. Therefore, in a fixed-base positional system, the arithmetic operation of multiplication by the base (or a power of the base) is strictly homomorphic to a physical leftward spatial shift of the numerical array. Conversely, a uniform rightward shift of the sequence by k bits corresponds perfectly to integer division (seeking the quotient) by 2^k, wherein the lower-order bits are either truncated or funneled into an adjacent remainder register[4].

Encoding Arithmetic in Orbital Angular Momentum Arrays

To implement this mathematical isomorphism optically, the discrete data bits must be reliably encoded into physical optical signals. OAM modes serve as the optimal information carriers due to their mutual orthogonality and highly distinct spatial intensity profiles. In the ODNN-based arithmetic unit, a logical "1" at a specific bit position is physically represented by the presence of an OAM beam generated at a precisely defined spatial coordinate on the transverse input plane. A logical "0" is represented by the absence of optical power at that coordinate.

If an optical input array consisting of spatially distributed OAM modes—often referred to as an OAM Shift Keying (OAM-SK) signal—is injected into the ODNN, the network's optimized phase screens are tasked with routing each incoming OAM beam to its adjacent spatial coordinate to the left[4]. By flawlessly executing this spatial shift without altering the topological charge, the ODNN optically computes a left-shift logic operation, outputting an OAM array that represents the multiplicand effectively multiplied by the operational base.

The profound advantage of this method lies in its massive physical parallelism. Regardless of

the bit-width or the complexity of the input integer, the spatial shift occurs simultaneously for all encoded bits as the light wavefront propagates through the physical diffractive layers[4]. Unlike sequential electronic Arithmetic Logic Units (ALUs) that suffer from carry propagation delays, the latency of the optical arithmetic unit is bounded entirely by the time of flight of photons traversing the physical distance of the ODNN—typically taking only fractions of a nanosecond.

Rotational Reconfiguration and Multi-ary Logic Expansion

A remarkable feature of the phase screens optimized for this architecture is their capacity for multi-functional reconfiguration through simple geometric manipulation. The operation type can be switched dynamically from multiplication to division without requiring a completely separate ODNN hardware module. By mechanically or electro-optically rotating the intermediate phase matrix by 90° clockwise, the diffractive properties of the screens are altered such that the resulting optical transfer function mirrors a rightward spatial shift[4]. Consequently, the input OAM array shifts to the right, enabling a computational architecture that seamlessly supports both fundamental functions using identical hardware[4].

Furthermore, the architecture is not restricted to binary logic systems. By expanding the semantic mapping of the spatial modes and utilizing multiple amplitude or phase levels, higher-radix computing can be achieved. In a 2-bit ternary configuration (base $N = 3$), the positional weights represent powers of 3 ($3^0, 3^1, 3^2$). When the ODNN executes a single-position spatial shift to the left on a ternary array, it effectively multiplies the input decimal value by 3[4]. The successful manipulation of ternary and quaternary data further highlights the unparalleled scalability of optical spatial shifting arithmetic compared to binary-locked electronic architectures.

Alternative Mechanisms: Geometric Coordinate Transformations

While fixed-base positional arithmetic via lateral spatial shifting offers immense advantages, alternative optical computing architectures manipulate the OAM mode itself to achieve arithmetic operations. These mechanisms rely on continuous phase gradient mappings to scale the topological charge directly, rather than shifting the physical position of the beam.

Coordinate transformations are heavily utilized for direct topological multiplication and division of OAM modes. The fundamental principle relies on azimuth-scaling and Fermat's spiral transformations to arbitrarily multiply or divide the topological charge integer[3]. The geometric coordinate mapping linking the input plane (r_1, θ_1) to the output plane (r_2, θ_2) is mathematically expressed as:

$$r_2 = c r_1^{1/n}, \quad \theta_2 = \theta_1 / n$$

where c denotes the radial scale parameter and n denotes the multiplicative-divisor parameter. To enforce this spiral transformation and achieve the multiplication or division of the OAM state, an unwrapping phase mask must be synthesized. The required phase function $Q(r_1, \theta_1)$ is derived as:

$$Q(r_1, \theta_1) = \frac{kd}{c r_1^q q} \cos(q\theta_1) - \frac{r_1^2}{2}$$

When $n = 2$, the system acts as a topological multiplier, doubling the intrinsic angular momentum of the photon. When $n = 1/2$, the system functions as a topological divider[3]. These operations are physically realized using vortex Dammann gratings, which apply weight factors to different diffraction orders based on a Fourier transform of a bladed optical grating. While Fermat's spiral transformations successfully modify the OAM topological charge, the fixed-base ODNN spatial shifting approach provides superior scalability for standard digital computing. Topological charge scaling requires increasingly complex phase unwrapping systems for large numbers, whereas the positional shifting architecture of the ODNN can process infinitely large numbers simply by increasing the transverse width of the phase matrix to accommodate more positional bits.

Physical System Architecture and Operating Parameters

Experimental Design and Optics

A comprehensive physical architecture for OAM fixed-base multiplication and division relies on precisely defined optical parameters. Contemporary frameworks, such as those formulated by Xinyue Tan and Debao Chen, utilize a Laguerre-Gaussian laser source operating at a near-infrared telecommunications wavelength of $\lambda = 1550$ nm[4]. The selection of the 1550 nm C-band is highly strategic; it ensures seamless compatibility with standard erbium-doped fiber amplifiers (EDFAs) and silicon-on-insulator (SOI) photonic waveguide architectures, thereby facilitating direct integration into existing global optical communication networks[4].

The input OAM beams are typically collimated with a tight waist radius of $w_0 = 0.35$ mm[4]. The ODNN itself is structured as a highly compact, multi-layer physical device, most commonly

modeled and fabricated as a three-layer or five-layer phase-only diffractive configuration[4].

ODNN Architectural Parameter	Specification / Value
Operational Wavelength (λ)	1550 nm[4]
Input Beam Waist Radius (w_0)	0.35 mm[4]
ODNN Phase Matrix Resolution	1200×1200 pixels[4]
Metasurface/SLM Pixel Dimensions	$8 \backslash \mathrm{mum} \times$ [cite: 4]
Total Transverse Propagation Distance	300 mm[4]
Diffractive Layers	3 to 5 layers[4]

The 1200×1200 matrix size paired with an $8 \backslash \mathrm{mum}$ pixel pitch yields a physical spatial layer extent of approximately $9.6 \text{ mm} \times 9.6 \text{ mm}$. This macroscopic area provides an ample spatial bandwidth product (SBWP) to support the independent modulation of multiple spatially distinct OAM modes[4]. The transverse free-space propagation distance between the input plane and the final output detection plane is generally established around 300 mm, with equidistant free-space gaps interleaving the phase layers to permit the diffractive mixing of complex wavefronts required for spatial routing[4].

OAM Comb Generation and Manipulation

In advanced high-throughput iterations of this technology, singular OAM modes are replaced by broadband OAM combs. An OAM comb is a multi-mode superposition state where the constituent topological charges are uniformly spaced, analogous to an optical frequency comb in the spectral domain[11]. The generation of high-uniformity OAM combs is achieved via sophisticated cross-phase modulation techniques, yielding an ideal comb containing topological charges ranging from $l = -20$ to $l = +20$ [11].

The ODNN architecture demonstrates exceptional proficiency in manipulating these complex comb states. Empirical models show that ODNNs can execute direct arithmetic operations on the entire OAM comb structure simultaneously. When tasked with arithmetic addition, subtraction, multiplication, and division of OAM combs, the diffractive architecture achieves extraordinary accuracy rates[11].

OAM Comb Operation	Empirical Accuracy
Addition	99.7%[11]
Subtraction	99.6%[11]
Multiplication	99.4%[11]
Division	99.7%[11]

These accuracy metrics—which maintain an average of 99.65% and never drop below a 99% threshold—underscore the immense capability of diffractive phase tailoring in managing dense, high-dimensional coherent optical states[11].

Perturbation Dynamics, Quantum Turbulence, and Error Tolerance

The transition of optical computing designs from idealized numerical simulations to physical hardware is often bottlenecked by extreme sensitivity to manufacturing tolerances, mechanical misalignment, and environmental phase noise. Evaluating the ODNN-based arithmetic unit involves rigorously scrutinizing its mode purity, diffraction efficiency, and robustness to physical and atmospheric perturbations.

Mode Purity and Diffraction Efficiency

Mode purity is the preeminent metric of success for an OAM-based computational system; if the spatial shifting operation severely distorts the helical phase front of the OAM beam, the topological charge l will degenerate, resulting in catastrophic data loss. The mode purity P is defined mathematically through the normalized inner product of the complex amplitude of the actual output field E_{out} and the ideal target Laguerre-Gaussian mode E_{target} :

$$P = \frac{\left| \iint E_{out}(x,y) E_{target}^{*}(x,y) dxdy \right|^2}{\iint |E_{out}(x,y)|^2 dxdy \iint |E_{target}(x,y)|^2 dxdy}$$

Simulations of the 3-layer ODNN arithmetic unit demonstrate spectacular preservation of the optical field, with average mode purities achieving between 95% and 99.65%, and the lowest isolated accuracy remaining above 91% even in worst-case cascaded logical scenarios[5]. The overall diffraction efficiency—which measures the proportion of input optical power successfully routed to the target spatial coordinates rather than scattered into background noise—similarly exceeds 95% under ideal operational conditions[5].

Robustness to Mechanical and Physical Misalignments

A physical optical neural network is susceptible to various spatial perturbations during assembly and operation. Three principal sources of hardware error must be analytically addressed: lateral displacement, rotation deviations, and phase quantization.

Lateral displacement arises from imperfect optomechanical alignment between the incident laser source, the input modulation plane, and the cascaded ODNN layers. The ODNN arithmetic unit exhibits a high degree of shift-invariance within tight spatial bounds. Minor lateral displacements on the sub-millimeter scale degrade mode purity slightly but generally maintain a sufficient signal-to-noise ratio for the OAM topological charge to be reliably decoded at the output plane[4].

Rotation deviations pose a more critical threat. Because the toggling of the arithmetic unit between multiplication and division relies on an exact 90° clockwise rotation of the phase matrix[4], angular deviations during mechanical or electro-optic rotation can severely corrupt the spatial routing path. The performance degradation curves indicate that while small angular misalignments of less than 1° are well-tolerated, larger deviations lead to a rapid breakdown of the spatial shifting matrix, dispersing the OAM beam across the output plane as unstructured speckle noise[4].

Environmental Phase Noise and Kolmogorov Turbulence

If the optical arithmetic unit is interfaced with Free-Space Optical (FSO) communication links to transmit encrypted computational data, the propagation environment becomes a severe factor. Atmospheric turbulence, driven by localized fluctuations in the refractive index of the medium, acts as an uncontrolled, stochastic diffractive layer that scrambles the delicately trained wavefronts[12].

In standard Kolmogorov turbulence theory, the parameter Ω denotes the turbulence strength, defined mathematically as a function of the beam size relative to the Fried parameter r_0 [12]. The Fried parameter represents the average correlation length of phase distortion. The probability $P(0)$ that a photon initially generated with an OAM topological charge of zero retains its zero state after propagating through a turbulent channel is derived as:

$$P(0) \approx [I_0(\beta) + I_1(\beta)] \exp(-\beta)$$

where $I_n(\cdot)$ is the modified Bessel function of the first kind, and $\beta = 1.8025\Omega^{5/3}$ [12]. When the turbulence strength $\Omega > 1$, the beam size exceeds the correlation length of the phase distortion. In this strong turbulence regime, the field undergoes significant spatial shifting and local deformations[12]. The spatial varying Bloch vectors $\mathbf{b}(\mathbf{r}) = (b_x, b_y, b_z)$,

which describe the localized state of the photon across its transverse profile, become scrambled. The Skyrmion number N, which quantifies the topological wrapping of the Bloch sphere, degrades rapidly, indicating modal crosstalk and catastrophic phase front distortions[12]. For an ODNN arithmetic architecture that relies precisely on spatial position to dictate the numeric value of a bit, an environmentally induced spatial shift is devastating. It can effectively scramble the positional data array, autonomously transforming a multiplication operation into a random numeric shift or division. To mitigate this vulnerability, OAM computational units must either be heavily confined within environmentally stabilized enclosures, coupled into specialized ring-core optical fibers that support stable OAM propagation, or employ rigorous topological phase recovery mechanisms prior to the detection plane[12].

Integration into All-Optical Logic and Networking Ecosystems

The utility of a fixed-base spatial shifting arithmetic unit is maximized when it is contextualized within a comprehensive, integrated all-optical processing ecosystem. Arithmetic manipulation is only one fraction of a generalized Arithmetic Logic Unit (ALU); logical gating and signal multiplexing are equally vital.

All-Optical Logical Gates and Adders

To complement physical multiplication and division, ODNNs have been successfully trained to execute fundamental logical boolean operations directly on OAM modes. Empirical research demonstrates that basic logic gates—including AND, OR, NOT, NAND, and NOR—can be mapped directly into single-model ODNN architectures[10].

In the implementation of an OAM logic AND gate, the presence of specific topological charges serves as the binary inputs. The diffractive network evaluates the physical presence of these incoming fields and outputs an OAM beam with a specific target topological charge if, and only if, both input modes are active[10]. The performance of these gates is exceptionally high; for instance, the ODNN accurately completes the logic OR operation with an output mode purity reaching 99.13%[10].

However, non-linear logic functions such as XOR and XNOR present a profoundly more complex computational challenge. The data distribution features in the training data for XOR are fundamentally "fuzzier" and lack simple linear separability. Consequently, a single passive linear ODNN layer cannot easily resolve XOR operations while maintaining high mode purity. To bypass this limitation, researchers physically cascade basic AND, OR, and NOT gates to construct XOR and XNOR networks, successfully achieving the requisite operational truth tables[10].

By unifying these cascaded logic gates, ODNN-based OAM Half-Adders have been successfully constructed. When the four fundamental input states ("00", "01", "10", "11") are sequentially fed into the diffractive half-adder, the system yields output mode purities greater than 91% for the SUM and CARRY operations[10].

Optical Logic Gate	Architectural Implementation	Achieved Mode Purity
OR Gate	Single ODNN Model	99.13%[10]
AND Gate	Single ODNN Model	> 95%[10]
XOR / XNOR Gates	Cascaded Basic Logic Units	Variable (High)[10]
Half-Adder	Cascaded Basic Logic Units	> 91%[10]

Optical Add-Drop Multiplexers (OADM) and Advanced Encryption

Moving vast quantities of computational data into and out of the arithmetic unit requires sophisticated, low-latency routing. ODNNs serve dual purposes as high-efficiency Optical Add-Drop Multiplexers (OADMs), capable of selectively coupling and separating distinct OAM modes for signal transport[5].

In a simulated four-channel communication link operating at 1550 nm, these OADMs demonstrate the ability to process multi-channel OAM signals carrying dense 16-Quadrature Amplitude Modulation (16-QAM) data payloads. The diffractive networks achieve a remarkably low Optical Signal-to-Noise Ratio (OSNR) penalty of approximately 1 dB at a stringent Bit Error Rate (BER) of 3.8×10^{-3} [5]. The integration of OADMs with spatial-shifting arithmetic units implies that a central optical processor can not only compute complex mathematical operations but also dynamically route the output integers into specific network channels based exclusively on their assigned topological charge. This facilitates seamless all-optical interconnections across data centers without the prohibitive latency of intermediate O-E-O conversions[5].

Furthermore, the infinite dimensionality of the OAM state space enables advanced hardware-level optical encryption. By leveraging the specific diffractive routing patterns of the arithmetic unit alongside spatial phase mapping, OAM mode logic can encrypt transmitted imagery and data. An encrypted image cannot be obtained or resolved correctly if incorrect OAM modes or mismatched multiplication/division transformations are applied during the physical decoding phase at the receiver plane, thereby guaranteeing the physical-layer security of information transmission during FSO communications[3].

Broader Implications and Non-Linear Expansions

While the fixed-base spatial shifting mechanics rely heavily on linear diffractive optics, theoretical expansions suggest integrating these paradigms with non-linear quantum substrates. Advanced optical traps formed by intersecting LG beams can confine Bose-Einstein Condensates (BECs) in highly structured optical potentials[14].

In these environments, the potential depth and harmonic confinement are dictated by the

optical power and waist of the intersecting OAM beams. The dynamics of the condensate are rigorously governed by the non-linear Gross-Pitaevskii Equation (GPE), which models the kinetic, potential, and non-linear interactions of the trapped matter-waves[14]. By integrating ODNN spatial shifting concepts with BEC optical trapping, researchers could theoretically execute spatial bit-shifts that rely not just on passive diffraction, but on the deterministic non-linear collision of quantum states, paving the way for ultra-low energy, quantum-coherent arithmetic logic units.

Conclusion

The application of Optical Diffractive Neural Networks for fixed-base digital arithmetic represents a watershed evolution in the architecture of all-optical computing. By mathematically redefining the fundamental operations of digital multiplication and division as uniform left and right geometric shifts of a positional bit sequence, and subsequently mapping these bitwise operations to the precise physical spatial shifting of discrete Orbital Angular Momentum arrays, researchers have successfully bypassed the fundamental latency and energy bottlenecks of electronic circuitry.

The primary findings of this paradigm underscore its immense viability. The computational elegance of executing left-shift multiplication and right-shift division operations using the identical three-layer ODNN hardware—toggled purely through a 90° clockwise physical rotation of the phase matrix—highlights a remarkable efficiency in component utilization. Operating at the industry-standard telecom wavelength of 1550 nm, the spatial shifting mechanism proves inherently dimension-agnostic, supporting not just binary arithmetic but extending naturally to ternary, quaternary, and broader high-radix computations without requiring substantial hardware overhauls.

Driven by the rigorous mathematical orthogonality of OAM topological charges, the diffractive process preserves mode purities up to an extraordinary 99.65%, maintaining the stringent error tolerances critical for reliable data processing. The demonstrated empirical robustness against lateral displacements, minor angular deviations, and extensive phase quantization further confirms its readiness for physical, solid-state implementation. In synergy with all-optical logical gates, OAM half-adders, and dynamic Optical Add-Drop Multiplexers, the fixed-base spatial shifting arithmetic unit provides the definitive quantitative core required to construct fully integrated, Turing-complete optical processors. As deep-learning methodologies for optimizing these diffractive matrices continue to advance, this architecture is uniquely positioned to redefine the speed, efficiency, and fundamental physical constraints of next-generation high-dimensional computing networks.

Works cited

1. (PDF) Optical computing - ResearchGate, https://www.researchgate.net/publication/317178854_Optical_computing
2. Optical angular momentum and atoms - PMC - NIH, https://pmc.ncbi.nlm.nih.gov/articles/PMC5247479/
3. Optical encryption via four operations of orbital angular momentum,

https://pubs.aip.org/aip/app/article/10/7/076113/3353357/Optical-encryption-via-four-operations-of-orbital
4. Optical diffractive neural network-based orbital angular momentum, https://opg.optica.org/oe/abstract.cfm?uri=oe-34-3-5277
5. Optical diffractive deep neural network-based orbital angular, https://opg.optica.org/abstract.cfm?uri=oe-29-22-36936
6. Orbital and Spin Angular Momentum | Bohrium, https://www.bohrium.com/en/sciencepedia/feynman/keyword/total_angular_momentum
7. Rotations and Angular momentum, https://www.bartol.udel.edu/~seckel/courses/phys_811/lecture%20notes/rotations.pdf
8. Schematic of spin separation and Pancharatnam-Berry (PB), https://www.researchgate.net/figure/Schematic-of-spin-separation-and-Pancharatnam-Berry-PB-arising-from-polarization_fig5_295173830
9. Orbital angular momentum beams demultiplexing using a hybrid, https://opg.optica.org/prj/abstract.cfm?uri=prj-13-12-B120
10. Orbital angular momentum mode logical operation using optical, https://opg.optica.org/abstract.cfm?uri=prj-9-10-2116
11. Versatile manipulation of orbital angular momentum combs enabled, https://opg.optica.org/oe/fulltext.cfm?uri=oe-33-4-6771
12. Topological robustness of orbital angular momentum entanglement, https://arxiv.org/html/2603.10618v1
13. Concept and principle of an orbital angular momentum (OAM)-based, https://www.researchgate.net/figure/Concept-and-principle-of-an-orbital-angular-momentum-OAM-based-underwater-wireless_fig1_320847387
14. Symmetry-Driven Multi-Soliton Dynamics in Bose–Einstein ... - MDPI, https://www.mdpi.com/2073-8994/17/4/582